



Manifold for Type 1 Diabetes

Pre-clinical risk stratification and intervention-window detection

Type 1 Diabetes has a years-long pre-clinical prodrome. Current risk-stratification tools (TrialNet DPTRS, Index60, M120) produce strong population-level survival curves but fail to identify **individual-trajectory inflection points** — the window within which a Stage-2-appropriate intervention delivers maximal benefit. **Manifold** is a causal-intelligence platform, configured for the T1D domain to detect these inflections: causal discovery on multi-omic cohorts, criticality estimators (DNB, Bayesian change-point, deep survival) feeding a composite fragility score (ΩF), and Pearl do-calculus with interdiction for principled therapy-selection framing.

Teplizumab's 2022 approval transformed the pre-clinical window from academic interest to actionable intervention. Each additional month of accurate per-patient stratification extends the period within which a Stage-2-appropriate intervention remains beneficial.

ΩF PILLAR MAPPING - T1D PHYSIOLOGY

ΩF PILLAR	T1D READING
I — Interaction	Autoantibody-to-autoantibody coupling (IAA, GADA, IA-2A, ZnT8); HLA × viral-trigger epistasis
R — Reserve	β -cell functional mass proxied by C-peptide AUC on MMTT; glucose-response amplitude
J — Junction	Immune-metabolic subsystem coupling (transfer entropy between immune-cell subsets and glycemic variability)
C — Cascade	Autoreactive T-cell clonal expansion; epitope spreading rate
T — Temporal	Hazard of next stage transition given current trajectory

PROPOSED PARTNERSHIP

Partnership with a TEDDY-affiliated principal investigator (or equivalent cohort steward — TrialNet, DiPiS, BabyDiab) is sought for:

1. Retrospective data access under the partnering institution's existing Data Use Agreement.
2. Clinical co-authorship on the benchmark manuscript (ΩF versus DPTRS; ten to twelve weeks from data access).
3. Co-application to Breakthrough T1D, the Helmsley Charitable Trust, or NIH NIDDK for the Phase 2 prospective arm.

Apex Analytica contributes the causal-inference engine, the ΩF framework, and the engineering team. Full brief: M-T1D-01 .